# TrES-2b

**R-1445**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_190622` · created 2026-08-01T09:04:15+00:00

**BENCHMARK: ACCURATE — fit consistent with published values**

## Results

| | |
|---|---|
| **Mid-Transit Time (Tmid)** | 2458656.741 +/- 0.0005 BJD_TDB |
| **Ratio of Planet to Stellar Radius (Rp/R*)** | 0.1159 +/- 0.0074 |
| **Transit depth (Rp/Rs)^2** | 1.34 +/- 0.17 [%] |
| **Orbital Inclination (inc)** | 85.0 +/- 0.56 |
| **Airmass coefficient 1 (a1)** | 1.278 +/- 0.011 |
| **Airmass coefficient 2 (a2)** | -0.2218 +/- 0.0054 |
| **Scatter in the residuals of the lightcurve fit is** | 0.86 % |
| **Best Comparison Star** | None |
| **Optimal Aperture** | 5.4 |
| **Optimal Annulus** | 8.55 |
| **Transit Duration (day)** | 0.0871 +/- 0.0062 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| **Rp/R■ fit vs published** | 0.1159 ± 0.0074 vs 0.1254 (deviation 0.87σ) |
| **Duration fit vs published** | 0.0871 d vs 0.0746 d (deviation 1.36σ) |
| **Mid-time O–C** | -1.7 min |
| **Depth SNR** | 10.6 |
| **Residual scatter** | 0.86 % |

## Light curve

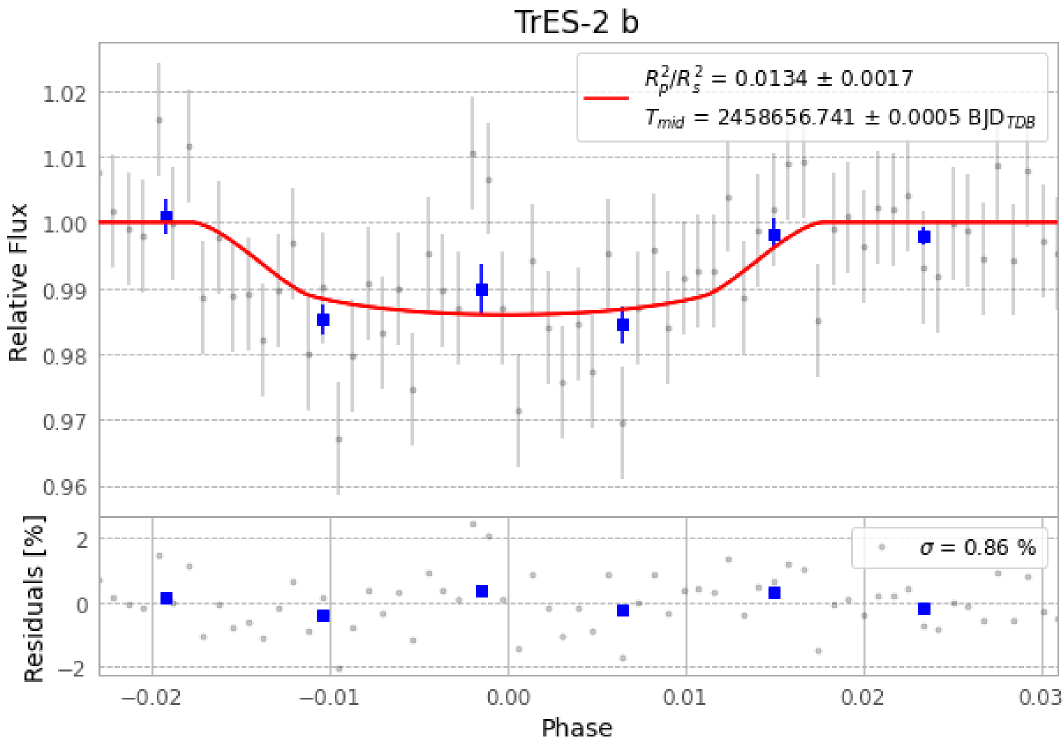


Figure 1 — FinalLightCurve_TrES-2 b_2019-06-21.png (SHA-256 83f1168b6856426d…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 100], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5dc1e7399f913307…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

## Data provenance

65 FITS frames (43 MB), `TRES-2190622042112.FITS` → `TRES-2190622073312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-190622.log (fa0738a1aa4b…), FinalLightCurve_TrES-2 b_2019-06-21.png (83f1168b6856…), FinalLightCurve_TrES-2 b_2019-06-21.csv (0541254f4459…)

## Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)